

A Prime-Power Obstruction for Equal Divisor Sums at Consecutive Integers

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Abstract

Let

$$\sigma(n) = \sum_{d|n} d$$

denote the sum-of-divisors function. We prove that if

$$\sigma(n) = \sigma(n+1) = S,$$

then S has at least two distinct prime divisors. Consequently, the common sum-of-divisors value of two consecutive positive integers cannot be a prime power. More precisely, if S is odd, then $3 \mid S$ and S has a prime divisor congruent to 1 modulo 3.

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1 Introduction

Let σ denote the sum-of-divisors function on $\mathbb{Z}^+$, defined by

$$\sigma(n) = \sum_{d|n} d.$$

We consider the equation

$$\sigma(n) = \sigma(n+1). \tag{1}$$

Questions concerning equal values of σ at consecutive arguments go back at least to Erdős [3]. Sierpiński explicitly asked whether (1) has infinitely many solutions [8, p. 166]; this remains open. Mąkowski [7] made an early study of equations involving equal values of the sum-of-divisors function. Guy and Shanks [6] later gave a constructed solution of (1) based on factorisation patterns observed among smaller solutions.

Erdős, Pomerance and Sárközy [4] proved that, for sufficiently large x , the number of solutions $n \leq x$ is at most

$$x \exp\left(-(\log x)^{1/3}\right).$$

A detailed computational study was carried out by Benito [2], who found 1268 solutions with

$$n \leq 1.5 \cdot 10^{10}.$$

Every common value occurring in that computation was divisible by 6. Benito also used the classical characterisation of integers with odd divisor sum to show that any odd common value in (1) must be divisible by 3.

Bayless and Kinlaw [1] subsequently obtained bounds for the sum of the reciprocals of the solutions.

The more general equation

$$\sigma(n) = \sigma(n + k)$$

has also been studied. Weingartner [9] proved, assuming Schinzel's Hypothesis H, that for every fixed even k there are infinitely many solutions. Yamada [10] obtained further results on the shifted equation, and Ford [5] proved unconditionally that it has infinitely many solutions for a positive proportion of positive integers k . These results do not settle the consecutive case $k = 1$.

The purpose of this paper is to prove the following restriction on the common value in (1).

Theorem 1.1. *Let $n \in \mathbb{Z}^+$ and suppose*

$$S = \sigma(n) = \sigma(n + 1).$$

Then S has at least two distinct prime divisors. In particular, there do not exist a prime P and a positive integer k such that

$$\sigma(n) = \sigma(n + 1) = P^k.$$

Moreover, if S is odd, then

$$3 \mid S$$

and there exists a prime $\ell \mid S$ such that

$$\ell \equiv 1 \pmod{3}.$$

The bound of two distinct prime divisors is best possible, since

$$\sigma(14) = \sigma(15) = 24 = 2^3 \cdot 3.$$

2 Preliminary lemmas

We use two standard elementary facts about the sum-of-divisors function.

Lemma 2.1. *If $\gcd(m, n) = 1$, then*

$$\sigma(mn) = \sigma(m)\sigma(n).$$

Proof. Every divisor of mn has a unique representation ab with $a \mid m$ and $b \mid n$. Hence

$$\sigma(mn) = \sum_{a \mid m} \sum_{b \mid n} ab = \left(\sum_{a \mid m} a \right) \left(\sum_{b \mid n} b \right) = \sigma(m)\sigma(n).$$

□

Lemma 2.2. *If $\sigma(m)$ is odd, then m is a square or twice a square.*

Proof. Write

$$m = 2^a \prod_{i=1}^r p_i^{b_i},$$

where the p_i are distinct odd primes. By Lemma 2.1,

$$\sigma(m) = \sigma(2^a) \prod_{i=1}^r \sigma(p_i^{b_i}).$$

The factor

$$\sigma(2^a) = 2^{a+1} - 1$$

is always odd. For an odd prime p_i ,

$$\sigma(p_i^{b_i}) = 1 + p_i + \cdots + p_i^{b_i}$$

is a sum of $b_i + 1$ odd terms, and hence is odd if and only if b_i is even.

Thus every exponent in the odd part of m is even. If a is even, then m is a square; if a is odd, then m is twice a square. $\square$

3 Proof of the main theorem

We divide the proof according to the parity of the common value S .

Case 1: S is even

Suppose

$$S = \sigma(n) = \sigma(n+1)$$

is even. Then

$$2 \mid S.$$

It therefore suffices to find an odd prime divisor of S .

Let m be the even member of $\{n, n+1\}$. Write

$$m = 2^a u,$$

where $a \geq 1$ and u is odd. By Lemma 2.1,

$$S = \sigma(m) = \sigma(2^a) \sigma(u) = (2^{a+1} - 1) \sigma(u).$$

Since $a \geq 1$,

$$2^{a+1} - 1 > 1,$$

and this integer is odd. Hence it has an odd prime divisor q , and therefore

$$q \mid S.$$

Together with $2 \mid S$, this shows that S has at least two distinct prime divisors.

Case 2: S is odd

Suppose instead that

$$S = \sigma(n) = \sigma(n+1)$$

is odd.

By Lemma 2.2, both n and $n+1$ are either squares or twice squares. Since n and $n+1$ are consecutive, one is odd and the other is even.

The odd member cannot be twice a square, so it must be a square. The even member cannot also be a square, since two positive squares cannot differ by 1: if

$$x^2 - y^2 = 1$$

for positive integers $x > y$, then

$$(x-y)(x+y) = 1,$$

which is impossible.

Thus the two members have the forms

$$x^2 \quad \text{and} \quad 2y^2,$$

where x is odd.

Let $2y^2$ be the even member. Write

$$y = 2^t u,$$

where $t \geq 0$ and u is odd. Then

$$2y^2 = 2^{2t+1} u^2.$$

Since

$$\gcd(2^{2t+1}, u^2) = 1,$$

Lemma 2.1 gives

$$S = \sigma(2y^2) = \sigma(2^{2t+1})\sigma(u^2).$$

Therefore

$$\sigma(2^{2t+1}) \mid S.$$

But

$$\sigma(2^{2t+1}) = 2^{2t+2} - 1.$$

Since

$$2^{2t+2} = 4^{t+1} \equiv 1 \pmod{3},$$

we obtain

$$3 \mid \sigma(2^{2t+1}),$$

and hence

$$3 \mid S. \tag{2}$$

This recovers the odd case of Benito's divisibility result [2, Theorem 2]. We now show that S possesses a second prime divisor, and in fact one congruent to 1 modulo 3.

The odd member is x^2 , so

$$S = \sigma(x^2).$$

By (2), we have $S > 1$, and hence $x > 1$. Write

$$x = \prod_{i=1}^r q_i^{v_i},$$

where the q_i are distinct primes and $v_i \geq 1$. By Lemma 2.1,

$$S = \prod_{i=1}^r \sigma(q_i^{2v_i}).$$

Since $3 \mid S$, at least one factor is divisible by 3. Thus for some prime $q = q_i$ and corresponding exponent $v = v_i$,

$$3 \mid \sigma(q^{2v}).$$

Now

$$\sigma(q^{2v}) = 1 + q + q^2 + \cdots + q^{2v}.$$

We consider q modulo 3.

If

$$q \equiv 0 \pmod{3},$$

then

$$\sigma(q^{2v}) \equiv 1 \pmod{3},$$

which is impossible.

If

$$q \equiv -1 \pmod{3},$$

then

$$\sigma(q^{2v}) \equiv 1 - 1 + 1 - \cdots + 1 \equiv 1 \pmod{3},$$

since there are $2v + 1$ terms. This is again impossible.

Therefore

$$q \equiv 1 \pmod{3}.$$

It follows that

$$\sigma(q^{2v}) \equiv 2v + 1 \pmod{3}.$$

Since $3 \mid \sigma(q^{2v})$,

$$3 \mid 2v + 1.$$

Write

$$2v + 1 = 3r$$

for some positive integer r . Then

$$\sigma(q^{2v}) = \frac{q^{2v+1} - 1}{q - 1} = \frac{q^{3r} - 1}{q - 1}.$$

Since

$$q^3 - 1 = (q - 1)(q^2 + q + 1)$$

and

$$q^3 - 1 \mid q^{3r} - 1,$$

we obtain

$$q^2 + q + 1 \mid \sigma(q^{2v}).$$

Consequently,

$$q^2 + q + 1 \mid S. \quad (3)$$

Since $q \equiv 1 \pmod{3}$, write

$$q = 3h + 1$$

for some positive integer h . Then

$$q^2 + q + 1 = 3(3h^2 + 3h + 1).$$

Moreover,

$$3h^2 + 3h + 1 \equiv 1 \pmod{3}$$

and

$$3h^2 + 3h + 1 > 1.$$

Hence $3h^2 + 3h + 1$ has some prime divisor $\ell \neq 3$. By (3),

$$\ell \mid S.$$

Thus S has at least the two distinct prime divisors 3 and ℓ .

It remains to determine the congruence class of ℓ . Since

$$\ell \mid q^2 + q + 1,$$

we have

$$q^3 \equiv 1 \pmod{\ell}.$$

Also $\ell \nmid q$, since otherwise

$$q^2 + q + 1 \equiv 1 \pmod{\ell},$$

contrary to $\ell \mid q^2 + q + 1$. Thus the multiplicative order of q modulo ℓ is well defined.

Furthermore,

$$q \not\equiv 1 \pmod{\ell},$$

because otherwise

$$q^2 + q + 1 \equiv 3 \equiv 0 \pmod{\ell},$$

which would imply $\ell = 3$, contrary to the choice of ℓ . Therefore the multiplicative order of q modulo ℓ is exactly 3. Since this order divides $\ell - 1$,

$$3 \mid \ell - 1.$$

Hence

$$\ell \equiv 1 \pmod{3}.$$

The theorem now follows in both cases. Therefore

$$\sigma(n) = \sigma(n + 1) = S$$

always implies that S has at least two distinct prime divisors. Consequently,

$$S \neq P^k$$

for every prime P and positive integer k .

4 Concluding remarks

The argument gives an elementary obstruction on the common value in

$$\sigma(n) = \sigma(n+1).$$

No such common value can be a prime power.

The result is best possible with respect to the number of distinct prime divisors, since

$$\sigma(14) = \sigma(15) = 24 = 2^3 \cdot 3.$$

The odd case yields additional information. Any hypothetical odd common value must be divisible by 3 and by a second prime congruent to 1 modulo 3. This is consistent with Benito's computational observation that every common value in her search was divisible by 6 [2], although the present argument does not rule out odd common values altogether.

For an even common value, the proof obtains an odd prime divisor from the factor

$$2^{a+1} - 1,$$

but gives no universal congruence condition on that prime. It would be interesting to determine whether information from both members of the consecutive pair can yield stronger universal restrictions on the prime factorisation of the common value.

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